Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **3 kg** block slides down a **30°** ramp with **8.7 N** friction up the ramp.
1. Use g=**9.8 m/s²**.
2. **Find along-ramp acceleration** and normal force.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$.
For a constant-acceleration interval. **Check signs**, units and what the equation treats as true; keep the original step visible when revising.

Worked reasoning: A student says that a car moving *east* must accelerate *east*. A car that moves *east* while braking is one example that shows this is wrong: its acceleration points *west*. One example can disprove a statement that says something is always true.

Your turn: 1. **Draw velocity** and acceleration arrows for that counterexample.
2. **Write a corrected statement about** their directions.

Car braking while moving east

East is right.

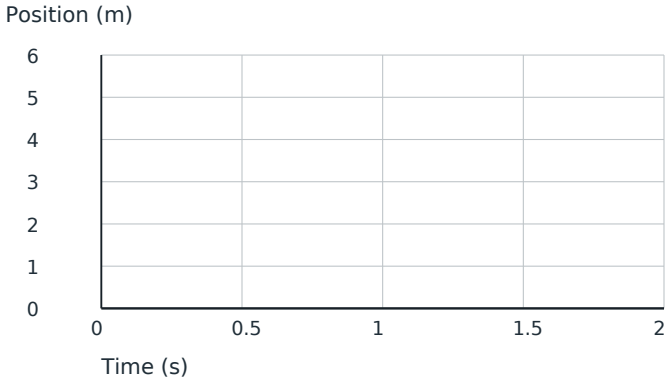
Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

03 Find the first bad step

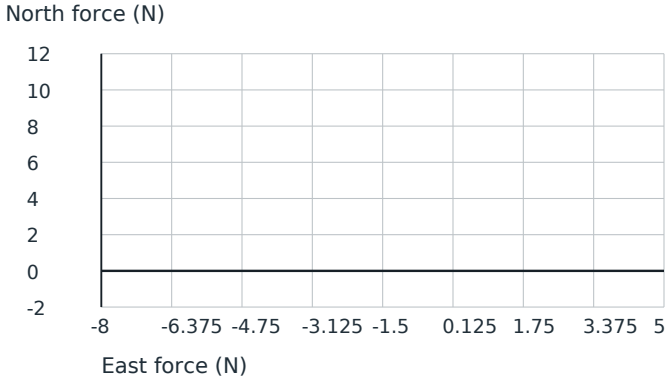
A robot starts at x = **5 m** and reaches x = **1 m** after **2 s**. A student calculates v = **1/2** = **0.5 m/s**.
1. **Sketch the motion** and correct the calculation.
2. **When would** dividing final position by elapsed time give the correct average velocity?



Marzano check - circle: ① ② ③

04 Check the axis

A rope exerts **10 N** at **30°** *west* of *north*. A student writes F_x = **10** cos **30°** and F_y = **10** sin **30°**. *East* is +x; *north* is +y. **Sketch the vector** and correct both the signs and trigonometric choices.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Fresh recheck

Check your corrected methods on two new cases.
(a) **Find the components of an 8 N** force at **30°** *south* of *east*.
(b) A **5 kg** cart has **7 N** right and **22 N** left. **Find its acceleration**. Does that alone establish its direction of motion?

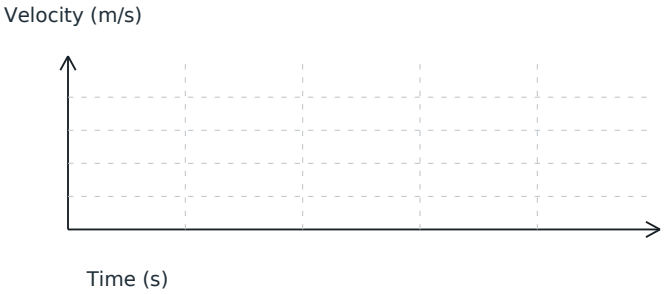


Case (b): mass 5 kg.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

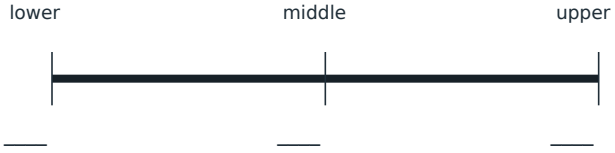
Velocity changes from **-5** to **-35 m/s** in **5 s**, then stays at **-35 m/s** for **9 s**. **Find acceleration** and **describe the speed trend** in each interval.
Sketch the velocity–time graph; **label the given values**.



Marzano check - circle: ① ② ③

07 Exit check

A robot starts at **3 ± 1 m** and moves at a constant **4 ± 0.5 m/s** for exactly **2 s**. Treat the input bounds independently.
1. **Find the central final position** and full interval.
2. **Name one specific error you** checked for.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A student calculates acceleration using the applied force alone and gets **5 m/s²** for a **2 kg** cart. The measured acceleration is **3 m/s²**. Infer a possible opposing force, then **name another explanation** to test.



2 kg cart: measured acceleration 3 m/s² right.

Marzano check - circle: ① ② ③